CSCA08H Worksheet: Arithmetic Operators

1. **Arithmetic Operators**

   Here are some excerpts from the official Python 3 documentation:

```
  x / y    quotient of x and y
  x // y   floored quotient of x and y  (1)
  x % y    remainder of x / y

(1) Also referred to as integer division. The resultant value is a whole integer, though the
    result's type is not necessarily int. The result is always rounded towards minus infinity:
    1 // 2 is 0, (-1) // 2 is -1, 1 // (-2) is -1, and (-1) // (-2) is 0.
...

Python fully supports mixed arithmetic: when a binary arithmetic operator has operands of
different numeric types, the operand with the "narrower" type is widened to that of the other,
where integer is narrower than floating point, which is narrower than complex.

...

Division of integers yields a float, while floor division of integers results in an integer.
```

   Let's make sure we understand how to read this. By using the above and without running the code, complete the table below by filling in the values that the expressions produce and the types of those values.

(a)

Python Expression	Result	Type of Result
`9 / 3`	`3.0`	`float`
`9 // 3`		

(b)

Python Expression	Result	Type of Result
`10 / 4`		
`10 // 4`		
`10 // 3`		
`10 % 3`		

2. **Arithmetic Operators**

   For which positive integers `n` does `n % 2` produce `0`? ____________________

   For which positive integers `n` does `n % 2` produce `1`? ____________________

3. **Order of Precedence of Arithmetic Operators**

Operator precedence of arithmetic operators in Python (from highest to lowest):

```
()              Parentheses
**              Exponent
+x, -x          Unary plus, Unary minus
*, /, //, %     Multiplication, Division, Floor/Integer division, Modulus
+, -            Addition, Subtraction
```

Using the above rules of precedence, fill in the table below by adding parentheses to indicate the order, in which the operations are evaluated.

Python Expression	**Python Expression Parenthesised**
`-2 + 4 * 7`	`((-2) + (4 * 7))`
`3 + 5 * 2`	
`4 + 8 / 2 ** 2 / -2`	